

MCS 2019

LITHIUM

U.S. Geological Survey, Mineral Commodity Summaries, January 2019

LITHIUM

(Data in metric tons of lithium content unless otherwise noted)

Domestic Production and Use: The only lithium production in the United States was from a brine operation in Nevada. Two companies produced a wide range of downstream lithium compounds in the United States from domestic or imported lithium carbonate, lithium chloride, and lithium hydroxide. Domestic production data were withheld to avoid disclosing company proprietary data.

Although lithium markets vary by location, global end-use markets are estimated as follows: batteries, 56%; ceramics and glass, 23%; lubricating greases, 6%; polymer production, 4%; continuous casting mold flux powders, 3%; air treatment, 2%; and other uses, 6%. Lithium consumption for batteries has increased significantly in recent years because rechargeable lithium batteries are used extensively in the growing market for portable electronic devices and increasingly are used in electric tools, electric vehicles, and grid storage applications. Lithium minerals were used directly as ore concentrates in ceramics and glass applications.

Salient Statistics—United States:	2014	2015	2016	2017	2018^e
	W	W	W	W	W
Production					
Imports for consumption	2,130	2,750	3,140	3,330	4,000
Exports	1,420	1,790	1,520	1,960	1,600
Consumption, estimated	12,000	12,000	13,000	13,000	13,000
Price, annual average, battery-grade lithium carbonate, dollars per metric ton ²	6,690	6,500	8,650	15,000	17,000
Employment, mine and mill, number	70	70	70	70	70
Net import reliance ³ as a percentage of estimated consumption	>25	>25	>50	>50	>50

Recycling: One domestic company has recycled lithium metal and lithium-ion batteries since 1992 at its facility in British Columbia, Canada. In 2015, the company began operating the first U.S. recycling facility for lithium-ion vehicle batteries in Lancaster, OH.

Import Sources (2014–17): Argentina, 51%; Chile, 44%; China, 3%; Russia, 1%; and other, 1%.

Tariff:	Item	Number	Normal Trade Relations
			12–31–18
	Other alkali metals	2805.19.9000	5.5% ad val.
	Lithium oxide and hydroxide	2825.20.0000	3.7% ad val.
	Lithium carbonate:		
	U.S. pharmaceutical grade	2836.91.0010	3.7% ad val.
	Other	2836.91.0050	3.7% ad val.

Depletion Allowance: 22% (Domestic), 14% (Foreign).

Government Stockpile:⁴

Material	Inventory As of 9–30–18	FY2018		FY 2019	
		Potential Acquisitions	Potential Disposals⁵	Potential Acquisitions	Potential Disposals⁵
Lithium cobalt oxide (kilograms, gross weight)	600	600	—	—	—
Lithium nickel cobalt aluminum oxide (kilograms, gross weight)	1,620	2,160	—	—	—
Lithium-ion precursors (kilograms, gross weight)	—	—	—	19,000	—

Events, Trends, and Issues: In May 2018, the U.S. Department of the Interior, in coordination with other executive branch agencies, published a list of 35 critical minerals (83 FR 23295), including lithium. This list was developed to serve as an initial focus, pursuant to Executive Order 13817, “A Federal Strategy to Ensure Secure and Reliable Supplies of Critical Minerals” (82 FR 60835).

Excluding U.S. production, worldwide lithium production in 2018 increased by 23% to 85,000 tons of lithium content from 69,000 tons of lithium content in 2017 in response to increased lithium demand for battery applications. This follows an increase of 74% in worldwide production in 2017 from that of 2016, owing primarily to a threefold increase in Australia’s spodumene production, including more than 11,000 tons of lithium content in direct shipping ore that

LITHIUM

was exported to China for processing. Consumption of lithium in 2018 was projected to be about 47,600 tons of lithium content, an increase of 20% from 39,700 tons of lithium content in 2017. Worldwide lithium production capacity was estimated to be 91,000 tons of lithium content per year.

Spot lithium carbonate prices in China decreased from approximately \$21,000 per ton at the beginning of the year to about \$12,000 per ton in the third quarter owing to worldwide lithium production exceeding worldwide lithium consumption. For large fixed contracts, however, Industrial Minerals reported an annual average U.S. lithium carbonate price of \$17,300 per metric ton in 2018, a 15% increase from that of 2017.

Five spodumene operations in Australia and two brine operations each in Argentina and Chile accounted for the majority of world lithium production. The leading spodumene operation in Australia increased its spodumene concentrate production by about 40% in 2018 and remained the world's largest lithium producer. Two new Australian spodumene operations ramped up production in 2017, and five additional spodumene operations ramped up production in 2018.

Lithium supply security has become a top priority for technology companies in the United States and Asia. Strategic alliances and joint ventures among technology companies and exploration companies continued to be established to ensure a reliable, diversified supply of lithium for battery suppliers and vehicle manufacturers. Brine operations were under development in Argentina, Bolivia, Chile, China, and the United States; pegmatite mining operations were under development in Australia, Austria, Canada, China, Czechia, Finland, Mali, Namibia, Portugal, Serbia, and Spain; and lithium-clay mining operations were under development in Mexico and the United States.

World Mine Production and Reserves: Reserves for Brazil, Chile, China, and Zimbabwe were revised based on new information from Government and industry sources.

	Mine production		Reserves ⁶
	2017	2018 ^e	
United States	W	W	35,000
Argentina	5,700	6,200	2,000,000
Australia	40,000	51,000	⁷ 2,700,000
Brazil	200	600	54,000
Chile	14,200	16,000	8,000,000
China	6,800	8,000	1,000,000
Portugal	800	800	60,000
Namibia	—	500	NA
Zimbabwe	800	1,600	70,000
World total (rounded)	⁸ 69,000	⁸ 85,000	14,000,000

World Resources: Owing to continuing exploration, lithium resources have increased substantially worldwide and total about 62 million tons. Identified lithium resources in the United States—from continental brines, geothermal brines, hectorite, oilfield brines, and pegmatites—are 6.8 million tons. Identified lithium resources in other countries have been revised to 55 million tons. Identified lithium resources in Argentina are 14.8 million tons; Bolivia, 9 million tons; Chile, 8.5 million tons; Australia, 7.7 million tons; China, 4.5 million tons; Canada, 2 million tons; Mexico, 1.7 million tons; Czechia, 1.3 million tons; Congo (Kinshasa), Russia, and Serbia, 1 million tons each; Zimbabwe, 540,000 tons; Mali and Spain, 400,000 tons each; Brazil and Germany, 180,000 tons each; Peru and Portugal, 130,000 tons each; Austria, 75,000 tons; Finland and Kazakhstan, 40,000 tons each; and Namibia, 9,000 tons.

Substitutes: Substitution for lithium compounds is possible in batteries, ceramics, greases, and manufactured glass. Examples are calcium, magnesium, mercury, and zinc as anode material in primary batteries; calcium and aluminum soaps as substitutes for stearates in greases; and sodic and potassic fluxes in ceramics and glass manufacture.

^eEstimated. W Withheld to avoid disclosing company proprietary data. NA Not available. — Zero.

¹Defined as production + imports – exports. Rounded to one significant digit to avoid disclosing company proprietary data.

²Source: Industrial Minerals, IM prices: Lithium carbonate, large contracts, delivered continental United States.

³Defined as imports – exports + adjustments for Government and industry stock changes.

⁴See Appendix B for definitions.

⁵Disposals are defined as any barter, rotation, sale, or upgrade of National Defense Stockpile stock.

⁶See Appendix C for resource and reserve definitions and information concerning data sources.

⁷For Australia, Joint Ore Reserves Committee-compliant reserves were about 1.4 million tons.

⁸Excludes U.S. production.

MCS 2020

LITHIUM

U.S. Geological Survey, Mineral Commodity Summaries, January 2020

LITHIUM

(Data in metric tons of lithium content unless otherwise noted)

Domestic Production and Use: The only lithium production in the United States was from a brine operation in Nevada. Two companies produced a wide range of downstream lithium compounds in the United States from domestic or imported lithium carbonate, lithium chloride, and lithium hydroxide. Domestic production data were withheld to avoid disclosing company proprietary data.

Although lithium markets vary by location, global end-use markets are estimated as follows: batteries, 65%; ceramics and glass, 18%; lubricating greases, 5%; polymer production, 3%; continuous casting mold flux powders, 3%; air treatment, 1%; and other uses, 5%. Lithium consumption for batteries has increased significantly in recent years because rechargeable lithium batteries are used extensively in the growing market for portable electronic devices and increasingly are used in electric tools, electric vehicles, and grid storage applications. Lithium minerals were used directly as ore concentrates in ceramics and glass applications.

Salient Statistics—United States:	2015	2016	2017	2018	2019^e
Production	W	W	W	W	W
Imports for consumption	2,750	3,140	3,330	3,420	2,500
Exports	1,790	1,520	1,960	1,660	1,700
Consumption, estimated ¹	2,000	3,000	3,000	3,000	2,000
Price, annual average, battery-grade lithium carbonate, dollars per metric ton ²	6,500	8,650	15,000	17,000	13,000
Employment, mine and mill, number	70	70	70	70	70
Net import reliance ³ as a percentage of estimated consumption	>25	>50	>50	>50	>25

Recycling: One domestic company has recycled lithium metal and lithium-ion batteries since 1992 at its facility in British Columbia, Canada. In 2015, the company began operating the first U.S. recycling facility for lithium-ion vehicle batteries in Lancaster, OH.

Import Sources (2015–18): Argentina, 53%; Chile, 40%; China, 3%; and other, 4%.

Tariff:	Item	Number	Normal Trade Relations 12–31–19
	Other alkali metals	2805.19.9000	5.5% ad val.
	Lithium oxide and hydroxide	2825.20.0000	3.7% ad val.
	Lithium carbonate:		
	U.S. pharmaceutical grade	2836.91.0010	3.7% ad val.
	Other	2836.91.0050	3.7% ad val.

Depletion Allowance: 22% (Domestic), 14% (Foreign).

Government Stockpile:⁴

Material	Inventory As of 9–30–19	FY 2019		FY 2020	
		Potential Acquisitions	Potential Disposals	Potential Acquisitions	Potential Disposals
Lithium cobalt oxide (kilograms, gross weight)	750	—	—	—	—
Lithium nickel cobalt aluminum oxide (kilograms, gross weight)	1,620	—	—	—	—
Lithium-ion precursors (kilograms, gross weight)	—	19,000	—	—	—

Events, Trends, and Issues: Excluding U.S. production, worldwide lithium production in 2019 decreased by 19% to 77,000 tons of lithium content from 95,000 tons of lithium content in 2018 in response to lithium production exceeding consumption and decreasing lithium prices. Global consumption of lithium in 2019 was estimated to be about 57,700 tons of lithium content, an increase of 18% from 49,100 tons of lithium content in 2018. However, consumption was lower than anticipated by the lithium industry owing to China scaling back subsidies on electric vehicles, consumers reducing lithium inventories, and lower electric vehicle sales volumes.

LITHIUM

Spot lithium carbonate prices in China decreased from approximately \$11,600 per ton at the beginning of the year to about \$7,300 per ton in December. For large fixed contracts, the annual average U.S. lithium carbonate price was \$13,000 per metric ton in 2019, a 24% decrease from that of 2018. Spot lithium hydroxide prices in China decreased from approximately \$15,500 per ton at the beginning of the year to about \$8,000 per ton in December. Spot lithium metal (99.9% Li) prices in China decreased from approximately \$120,000 per ton at the beginning of the year to about \$82,000 per ton in December.

Six mineral operations in Australia, two brine operations each in Argentina and Chile, and one brine and one mineral operation in China accounted for the majority of world lithium production. Owing to overproduction and decreased prices, several established lithium operations postponed capacity expansion plans. Junior mining operations in Australia, Canada, and Namibia ceased production altogether.

Lithium supply security has become a top priority for technology companies in the United States and Asia. Strategic alliances and joint ventures among technology companies and exploration companies continued to be established to ensure a reliable, diversified supply of lithium for battery suppliers and vehicle manufacturers. Brine-based lithium sources were in various stages of development in Argentina, Bolivia, Chile, China, and the United States; mineral-based lithium sources were in various stages of development in Australia, Austria, Brazil, Canada, China, Congo (Kinshasa), Czechia, Finland, Germany, Mali, Namibia, Portugal, Serbia, Spain, and Zimbabwe; and lithium-clay sources were in various stages of development in Mexico and the United States.

World Mine Production and Reserves: Reserves for Argentina, Australia, Brazil, Chile, the United States, and Zimbabwe were revised based on new information from Government and industry sources.

	Mine production		Reserves ⁵
	2018	2019 ^e	
United States	W	W	630,000
Argentina	6,400	6,400	1,700,000
Australia	58,800	42,000	⁶ 2,800,000
Brazil	300	300	95,000
Canada	2,400	200	370,000
Chile	17,000	18,000	8,600,000
China	7,100	7,500	1,000,000
Namibia	500	—	NA
Portugal	800	1,200	60,000
Zimbabwe	1,600	1,600	230,000
Other ⁷	—	—	<u>1,100,000</u>
World total (rounded)	⁸ 95,000	⁸ 77,000	17,000,000

World Resources: Owing to continuing exploration, identified lithium resources have increased substantially worldwide and total about 80 million tons. Lithium resources in the United States—from continental brines, geothermal brines, hectorite, oilfield brines, and pegmatites—are 6.8 million tons. Lithium resources in other countries have been revised to 73 million tons. Lithium resources, in descending order, are: Bolivia, 21 million tons; Argentina, 17 million tons; Chile, 9 million tons; Australia, 6.3 million tons; China, 4.5 million tons; Congo (Kinshasa), 3 million tons; Germany, 2.5 million tons; Canada and Mexico, 1.7 million tons each; Czechia, 1.3 million tons; Mali, Russia, and Serbia, 1 million tons each; Zimbabwe, 540,000 tons; Brazil, 400,000 tons; Spain, 300,000 tons; Portugal, 250,000 tons; Peru, 130,000 tons; Austria, Finland and Kazakhstan, 50,000 tons each; and Namibia, 9,000 tons.

Substitutes: Substitution for lithium compounds is possible in batteries, ceramics, greases, and manufactured glass. Examples are calcium, magnesium, mercury, and zinc as anode material in primary batteries; calcium and aluminum soaps as substitutes for stearates in greases; and sodic and potassic fluxes in ceramics and glass manufacture.

^eEstimated. W Withheld to avoid disclosing company proprietary data. NA Not available. — Zero.

¹Defined as production + imports – exports. Rounded to one significant digit to avoid disclosing company proprietary data.

²Source: Industrial Minerals, IM prices: Lithium carbonate, large contracts, delivered continental United States.

³Defined as imports – exports + adjustments for Government and industry stock changes.

⁴See Appendix B for definitions.

⁵See Appendix C for resource and reserve definitions and information concerning data sources.

⁶For Australia, Joint Ore Reserves Committee-compliant reserves were 1.7 million tons.

⁷Other countries with reported reserves include Finland, Mali, and Mexico.

⁸Excludes U.S. production.

MCS 2021

LITHIUM

U.S. Geological Survey, Mineral Commodity Summaries, January 2021

LITHIUM

(Data in metric tons of lithium content unless otherwise noted)

Domestic Production and Use: The only lithium production in the United States was from a brine operation in Nevada. Two companies produced a wide range of downstream lithium compounds in the United States from domestic or imported lithium carbonate, lithium chloride, and lithium hydroxide. Domestic production data were withheld to avoid disclosing company proprietary data.

Although lithium markets vary by location, global end-use markets are estimated as follows: batteries, 71%; ceramics and glass, 14%; lubricating greases, 4%; continuous casting mold flux powders, 2%; polymer production, 2%; air treatment, 1%; and other uses, 6%. Lithium consumption for batteries has increased significantly in recent years because rechargeable lithium batteries are used extensively in the growing market for portable electronic devices and increasingly are used in electric tools, electric vehicles, and grid storage applications. Lithium minerals were used directly as ore concentrates in ceramics and glass applications.

Salient Statistics—United States:

	<u>2016</u>	<u>2017</u>	<u>2018</u>	<u>2019</u>	<u>2020^e</u>
Production	W	W	W	W	W
Imports for consumption	3,140	3,330	3,420	2,620	2,900
Exports	1,520	1,960	1,660	1,680	1,400
Consumption, estimated ¹	3,000	3,000	3,000	2,000	2,000
Price, annual average, battery-grade lithium carbonate, dollars per metric ton ²	8,650	15,000	17,000	12,700	8,000
Employment, mine and mill, number	70	70	70	70	70
Net import reliance ³ as a percentage of estimated consumption	>50	>50	>50	>25	>50

Recycling: One domestic company has recycled lithium metal and lithium-ion batteries since 1992 at its facility in British Columbia, Canada. In 2015, the company began operating the first U.S. recycling facility for lithium-ion vehicle batteries in Lancaster, OH. Seven other companies located in Canada and the United States have begun recycling, or intend to begin recycling, lithium metal and lithium-ion batteries to some degree.

Import Sources (2016–19): Argentina, 55%; Chile, 36%; China, 5%; Russia, 2%; and other, 2%.

<u>Tariff:</u>	<u>Item</u>	<u>Number</u>	<u>Normal Trade Relations</u> <u>12–31–20</u>
	Other alkali metals	2805.19.9000	5.5% ad val.
	Lithium oxide and hydroxide	2825.20.0000	3.7% ad val.
	Lithium carbonate:		
	U.S. pharmaceutical grade	2836.91.0010	3.7% ad val.
	Other	2836.91.0050	3.7% ad val.

Depletion Allowance: 22% (domestic), 14% (foreign).

Government Stockpile:⁴

<u>Material</u>	<u>Inventory</u> <u>as of 9–30–20</u>	<u>FY 2020</u>		<u>FY 2021</u>	
		<u>Potential</u> <u>acquisitions</u>	<u>Potential</u> <u>disposals</u>	<u>Potential</u> <u>acquisitions</u>	<u>Potential</u> <u>disposals</u>
Lithium cobalt oxide (kilograms, gross weight)	750	—	—	—	—
Lithium nickel cobalt aluminum oxide (kilograms, gross weight)	2,700	—	—	—	—
Lithium-ion precursors (kilograms, gross weight)	—	—	—	—	—

Events, Trends, and Issues: Excluding U.S. production, worldwide lithium production in 2020 decreased by 5% to 82,000 tons of lithium content from 86,000 tons of lithium content in 2019 in response to lithium production exceeding consumption and decreasing lithium prices. Global consumption of lithium in 2020 was estimated to be 56,000 tons of lithium content, about the same as that of 2019. During the first half of 2020, the economic impact of the global COVID-19 pandemic was reported to have been a substantial factor in the reduction of customer demand. The second half of 2020 saw lithium demand increase owing primarily to strong growth in the lithium-ion battery market.

LITHIUM

Spot lithium carbonate prices in China decreased from approximately \$7,100 per ton at the beginning of the year to about \$6,200 per ton in November. For large fixed contracts, the annual average U.S. lithium carbonate price was \$8,000 per metric ton in 2020, a 37% decrease from that of 2019. Spot lithium hydroxide prices in China decreased from approximately \$7,800 per ton at the beginning of the year to about \$7,000 per ton in November. Spot lithium metal (99.9% lithium) prices in China decreased from approximately \$83,000 per ton at the beginning of the year to about \$71,000 per ton in November.

Five mineral operations in Australia, two brine operations each in Argentina and Chile, and two brine and one mineral operation in China accounted for the majority of world lithium production. Owing to overproduction and decreased prices, several established lithium operations postponed capacity expansion plans. Junior mining operations in Australia and Canada ceased production altogether.

Lithium supply security has become a top priority for technology companies in the United States and Asia. Strategic alliances and joint ventures among technology companies and exploration companies continued to be established to ensure a reliable, diversified supply of lithium for battery suppliers and vehicle manufacturers. Brine-based lithium sources were in various stages of development in Argentina, Bolivia, Chile, China, and the United States; mineral-based lithium sources were in various stages of development in Australia, Austria, Brazil, Canada, China, Congo (Kinshasa), Czechia, Finland, Germany, Mali, Namibia, Peru, Portugal, Serbia, Spain, and Zimbabwe; and lithium-clay sources were in various stages of development in Mexico and the United States.

World Mine Production and Reserves: Reserves for Argentina, Australia, Canada, Chile, China, the United States, Zimbabwe, and other countries were revised based on new information from Government and industry sources.

	Mine production		Reserves ⁵
	2019	2020 ^e	
United States	W	W	750,000
Argentina	6,300	6,200	1,900,000
Australia	45,000	40,000	⁶ 4,700,000
Brazil	2,400	1,900	95,000
Canada	200	—	530,000
Chile	19,300	18,000	9,200,000
China	10,800	14,000	1,500,000
Portugal	900	900	60,000
Zimbabwe	1,200	1,200	220,000
Other countries ⁷	—	—	2,100,000
World total (rounded)	⁸ 86,000	⁸ 82,000	21,000,000

World Resources:⁵ Owing to continuing exploration, identified lithium resources have increased substantially worldwide and total about 86 million tons. Lithium resources in the United States—from continental brines, geothermal brines, hectorite, oilfield brines, and pegmatites—are 7.9 million tons. Lithium resources in other countries have been revised to 78 million tons. Lithium resources are Bolivia, 21 million tons; Argentina, 19.3 million tons; Chile, 9.6 million tons; Australia, 6.4 million tons; China, 5.1 million tons; Congo (Kinshasa), 3 million tons; Canada, 2.9 million tons; Germany, 2.7 million tons; Mexico, 1.7 million tons; Czechia, 1.3 million tons; Serbia, 1.2 million tons; Peru, 880,000 tons; Mali, 700,000 tons; Zimbabwe, 500,000 tons; Brazil, 470,000 tons; Spain, 300,000 tons; Portugal, 270,000 tons; Ghana, 90,000 tons; and Austria, Finland, Kazakhstan, and Namibia, 50,000 tons each.

Substitutes: Substitution for lithium compounds is possible in batteries, ceramics, greases, and manufactured glass. Examples are calcium, magnesium, mercury, and zinc as anode material in primary batteries; calcium and aluminum soaps as substitutes for stearates in greases; and sodic and potassic fluxes in ceramics and glass manufacture.

^eEstimated. W Withheld to avoid disclosing company proprietary data. — Zero.

¹Defined as production + imports – exports + adjustments for Government and industry stock changes. Rounded to one significant digit to avoid disclosing company proprietary data.

²Source: Industrial Minerals (Fastmarkets IM), lithium carbonate, large contracts, delivered continental United States.

³Defined as imports – exports + adjustments for Government and industry stock changes.

⁴See Appendix B for definitions.

⁵See Appendix C for resource and reserve definitions and information concerning data sources.

⁶For Australia, Joint Ore Reserves Committee-compliant reserves were 2.8 million tons.

⁷Other countries with reported reserves include Austria, Congo (Kinshasa), Czechia, Finland, Germany, Mali, and Mexico.

⁸Excludes U.S. production.

MCS 2022

LITHIUM

U.S. Geological Survey, Mineral Commodity Summaries, January 2022

LITHIUM

(Data in metric tons of contained lithium unless otherwise noted)

Domestic Production and Use: The only lithium production in the United States was from one brine operation in Nevada. Two companies produced a wide range of downstream lithium compounds in the United States from domestic or imported lithium carbonate, lithium chloride, and lithium hydroxide. Domestic production data were withheld to avoid disclosing company proprietary data.

Although lithium markets vary by location, global end-use markets are estimated as follows: batteries, 74%; ceramics and glass, 14%; lubricating greases, 3%; continuous casting mold flux powders, 2%; polymer production, 2%; air treatment, 1%; and other uses, 4%. Lithium consumption for batteries has increased significantly in recent years because rechargeable lithium batteries are used extensively in the growing market for electric vehicles and portable electronic devices, and increasingly are used in electric tools, and grid storage applications. Lithium minerals were used directly as ore concentrates in ceramics and glass applications.

Salient Statistics—United States:

	<u>2017</u>	<u>2018</u>	<u>2019</u>	<u>2020</u>	<u>2021^e</u>
Production	W	W	W	W	W
Imports for consumption	3,330	3,420	2,620	2,460	2,500
Exports	1,960	1,660	1,660	1,170	1,900
Consumption, estimated ¹	3,000	3,000	2,000	2,000	2,000
Price, annual average, battery-grade lithium carbonate, dollars per metric ton ²	15,000	17,000	12,700	8,000	17,000
Employment, mine and mill, number	70	70	70	70	70
Net import reliance ³ as a percentage of estimated consumption	>50	>50	>25	>50	>25

Recycling: One domestic company has recycled lithium metal and lithium-ion batteries since 1992 at its facility in British Columbia, Canada. In 2015, the company began operating the first U.S. recycling facility for lithium-ion vehicle batteries in Lancaster, OH. About 25 companies in North America and Europe recycle lithium batteries or plan to do so. Partnerships between automobile companies and battery recyclers have been made to supply the automobile industry with a source of battery materials.

Import Sources (2017–20): Argentina, 54%; Chile, 37%; China, 5%; Russia, 3%; and other, 1%.

<u>Tariff:</u>	<u>Item</u>	<u>Number</u>	<u>Normal Trade Relations</u> <u>12–31–21</u>
	Lithium oxide and hydroxide	2825.20.0000	3.7% ad valorem.
	Lithium carbonate:		
	U.S. pharmaceutical grade	2836.91.0010	3.7% ad valorem.
	Other	2836.91.0050	3.7% ad valorem.

Depletion Allowance: 22% (domestic), 14% (foreign).

Government Stockpile:^{4, 5}

		<u>FY 2021</u>		<u>FY 2022</u>	
<u>Material</u>	<u>Inventory</u> <u>as of 9–30–21</u>	<u>Potential</u> <u>acquisitions</u>	<u>Potential</u> <u>disposals</u>	<u>Potential</u> <u>acquisitions</u>	<u>Potential</u> <u>disposals</u>
Lithium-cobalt oxide	752	—	—	—	—
Lithium-nickel-cobalt-aluminum oxide	2,698	—	—	—	—

Events, Trends, and Issues: Excluding U.S. production, worldwide lithium production in 2021 increased by 21% to approximately 100,000 tons from 82,500 tons in 2020 in response to strong demand from the lithium-ion battery market and increased prices of lithium. Global consumption of lithium in 2021 was estimated to be 93,000 tons, a 33% increase from 70,000 tons in 2020.

Spot lithium carbonate prices in China (cost, insurance, and freight [c.i.f.] North Asia) increased from approximately \$7,000 per ton in January to about \$26,200 per ton in November. For fixed contracts, the annual average U.S. lithium carbonate price was \$17,000 per ton in 2021, more than double that in 2020. Spot lithium hydroxide prices in China (c.i.f. North Asia) increased from approximately \$9,000 per ton in January to about \$27,400 per ton in November. Spot spodumene (6% lithium oxide) prices in China (c.i.f. China) increased from approximately \$450 per ton in January to about \$2,300 per ton in November. Spot lithium metal (99.9% lithium) prices in China increased from approximately \$77,000 per ton in January to about \$97,000 per ton in July.

Prepared by **Brian W. Jaskula** [(703) 648–4908, bjaskula@usgs.gov]

LITHIUM

Four mineral operations in Australia, two brine operations each in Argentina and Chile, and two brine and one mineral operation in China accounted for the majority of world lithium production. Additionally, smaller operations in Brazil, China, Portugal, the United States, and Zimbabwe also contributed to world lithium production. Owing to the resurgence in demand and increased prices of lithium in 2021, established lithium operations worldwide resumed capacity expansion plans which were postponed in 2020 in response to the global COVID-19 pandemic.

Lithium supply security has become a top priority for technology companies in Asia, Europe, and the United States. Strategic alliances and joint ventures among technology companies and exploration companies continued to be established to ensure a reliable, diversified supply of lithium for battery suppliers and vehicle manufacturers. Brine-based lithium sources were in various stages of development in Argentina, Bolivia, Chile, China, and the United States; mineral-based lithium sources were in various stages of development in Australia, Austria, Brazil, Canada, China, Congo (Kinshasa), Czechia, Finland, Germany, Mali, Namibia, Peru, Portugal, Serbia, Spain, the United States, and Zimbabwe; lithium-clay sources were in various stages of development in Mexico and the United States; and a searlesite source was in development in the United States.

World Mine Production and Reserves: Reserves for Argentina, Australia, and “Other countries” were revised based on new information from Government and industry sources.

	Mine production		Reserves ⁶
	2020	2021 ^e	
United States	W	W	750,000
Argentina	5,900	6,200	2,200,000
Australia	39,700	55,000	⁷ 5,700,000
Brazil	1,420	1,500	95,000
Chile	21,500	26,000	9,200,000
China	13,300	14,000	1,500,000
Portugal	348	900	60,000
Zimbabwe	417	1,200	220,000
Other countries ⁸	—	—	⁹ 2,700,000
World total (rounded)	⁹ 82,500	⁹ 100,000	22,000,000

World Resources:⁶ Owing to continuing exploration, identified lithium resources have increased substantially worldwide and total about 89 million tons. Identified lithium resources in the United States—from continental brines, geothermal brines, hectorite, oilfield brines, pegmatites, and searlesite—are 9.1 million tons. Identified lithium resources in other countries have been revised to 80 million tons. Identified lithium resources are distributed as follows: Bolivia, 21 million tons; Argentina, 19 million tons; Chile, 9.8 million tons; Australia, 7.3 million tons; China, 5.1 million tons; Congo (Kinshasa), 3 million tons; Canada, 2.9 million tons; Germany, 2.7 million tons; Mexico, 1.7 million tons; Czechia, 1.3 million tons; Serbia, 1.2 million tons; Russia, 1 million tons; Peru, 880,000 tons; Mali, 700,000 tons; Zimbabwe, 500,000 tons; Brazil, 470,000 tons; Spain, 300,000 tons; Portugal, 270,000 tons; Ghana, 130,000 tons; Austria, 60,000 tons; and Finland, Kazakhstan, and Namibia, 50,000 tons each.

Substitutes: Substitution for lithium compounds is possible in batteries, ceramics, greases, and manufactured glass. Examples are calcium, magnesium, mercury, and zinc as anode material in primary batteries; calcium and aluminum soaps as substitutes for stearates in greases; and sodic and potassic fluxes in ceramics and glass manufacture.

^eEstimated. W Withheld to avoid disclosing company proprietary data. — Zero.

¹Defined as production + imports – exports + adjustments for Government and industry stock changes. Rounded to one significant digit to avoid disclosing company proprietary data.

²Lithium carbonate, contract price, delivered Europe and United States. Source: Fastmarkets IM.

³Defined as imports – exports + adjustments for Government and industry stock changes.

⁴See Appendix B for definitions.

⁵Units are kilograms, gross weight.

⁶See Appendix C for resource and reserve definitions and information concerning data sources.

⁷For Australia, Joint Ore Reserves Committee or equivalent reserves were 3.8 million tons.

⁸Other countries with reported reserves include Austria, Canada, Congo (Kinshasa), Czechia, Finland, Germany, Mali, Mexico, and Serbia.

⁹Excludes U.S. production.

MCS 2023

LITHIUM

U.S. Geological Survey, Mineral Commodity Summaries, January 2023

LITHIUM

(Data in metric tons of contained lithium unless otherwise noted)

Domestic Production and Use: Commercial-scale lithium production in the United States was from one continental brine operation in Nevada. Lithium was also commercially produced from the brine-sourced waste tailings of a Utah-based magnesium producer. Two companies produced a wide range of downstream lithium compounds in the United States from domestic or imported lithium carbonate, lithium chloride, and lithium hydroxide. Domestic production data were withheld to avoid disclosing company proprietary data.

Although lithium markets vary by location, global end-use markets were estimated as follows: batteries, 80%; ceramics and glass, 7%; lubricating greases, 4%; continuous casting mold flux powders, 2%; air treatment, 1%; medical, 1%; and other uses, 5%. Lithium consumption for batteries increased significantly in recent years because rechargeable lithium batteries have been used extensively in the growing market for electric vehicles and portable electronic devices, and increasingly have been used in electric tools, and grid storage applications. Lithium minerals were used directly as ore concentrates in ceramics and glass applications.

Salient Statistics—United States:	2018	2019	2020	2021	2022^e
Production	W	W	W	W	W
Imports for consumption	3,420	2,620	2,460	2,640	3,400
Exports	1,660	1,660	1,200	1,870	2,700
Consumption, estimated ¹	3,000	2,000	2,000	2,000	3,000
Price, annual average-nominal, battery-grade lithium carbonate, dollars per metric ton ²	16,000	11,700	8,400	12,600	37,000
Employment, mine and mill, number	70	70	70	70	70
Net import reliance ³ as a percentage of estimated consumption	>50	>25	>50	>25	>25

Recycling: Construction of lithium battery recycling plants increased at a rapid pace. As of November 2022, about 44 companies in Canada and the United States and 47 companies in Europe recycled lithium batteries or planned to do so. Automobile companies and battery recyclers partnered to supply the automobile industry with a source of battery materials.

Import Sources (2018–21): Argentina, 51%; Chile, 40%; China, 4%; Russia, 3%; and other, 2%.

Tariff:	Item	Number	Normal Trade Relations 12–31–22
	Lithium oxide and hydroxide	2825.20.0000	3.7% ad valorem.
	Lithium carbonate:		
	U.S. pharmaceutical grade	2836.91.0010	3.7% ad valorem.
	Other	2836.91.0050	3.7% ad valorem.

Depletion Allowance: 22% (domestic), 14% (foreign).

Government Stockpile:^{4, 5}

		FY 2022		FY 2023	
Material	Inventory as of 9–30–22	Potential acquisitions	Potential disposals	Potential acquisitions	Potential disposals
Lithium-cobalt oxide	752	—	—	—	—
Lithium-nickel-cobalt-aluminum oxide	2,698	—	—	—	—

Events, Trends, and Issues: Excluding U.S. production, worldwide lithium production in 2022 increased by 21% to approximately 130,000 tons from 107,000 tons in 2021 in response to strong demand from the lithium-ion battery market and increased prices of lithium. Global consumption of lithium in 2022 was estimated to be 134,000 tons, a 41% increase from 95,000 tons in 2021.

Spot lithium carbonate prices in China (cost, insurance, and freight [c.i.f.]) increased from approximately \$35,000 per ton in January to about \$67,000 per ton in November. For fixed contracts, the annual average U.S. lithium carbonate price was \$37,000 per ton in 2022, almost three times higher than that in 2021. Spot lithium hydroxide prices in China (free on board) increased from approximately \$35,300 per ton in January to about \$78,000 per ton in November. Spot spodumene (6% lithium oxide) prices in China (c.i.f.) increased from approximately \$4,900 per ton in January to about \$5,800 per ton in November.

LITHIUM

Six mineral operations in Australia, one mineral tailings operation in Brazil, two brine operations each in Argentina and Chile, and three mineral and two brine operations in China accounted for the majority of world lithium production. Additionally, smaller operations in Brazil, Canada, China, Portugal, the United States, and Zimbabwe also contributed to world lithium production. Owing to the rapid increase in demand and prices of lithium in 2022, established lithium operations worldwide increased or were in the process of increasing production capacity.

The U.S. Department of Energy selected 12 lithium-based projects funded with \$1.6 billion from the 2022 U.S. Bipartisan Infrastructure Law to support new commercial-scale domestic facilities to extract and process lithium, manufacture battery components, recycle batteries, and develop new technologies to increase U.S. lithium reserves.

Lithium supply security has become a top priority for technology companies in Asia, Europe, and North America. Strategic alliances and joint ventures among technology companies and exploration companies continued to be established to ensure a reliable, diversified supply of lithium for battery suppliers and vehicle manufacturers. Brine-based lithium sources were in various stages of development or exploration in Argentina, Bolivia, Chile, China, and the United States; mineral-based lithium sources were in various stages of development or exploration in Australia, Austria, Brazil, Canada, China, Congo (Kinshasa), Czechia, Ethiopia, Finland, Germany, Ghana, Kazakhstan, Mali, Namibia, Nigeria, Peru, Portugal, Russia, Serbia, Spain, Thailand, the United States, and Zimbabwe; lithium-clay sources were in various stages of development or exploration in Mexico and the United States.

World Mine Production and Reserves: Reserves for Argentina, Australia, Brazil, Canada, Chile, China, the United States, Zimbabwe, and “Other countries” were revised based on information from company and Government reports.

	Mine production		Reserves ⁶
	2021	2022 ^e	
United States	W	W	1,000,000
Argentina	5,970	6,200	2,700,000
Australia	55,300	61,000	⁷ 6,200,000
Brazil	^e 1,700	2,200	250,000
Canada	—	500	930,000
Chile	28,300	39,000	9,300,000
China	^e 14,000	19,000	2,000,000
Portugal	^e 900	600	60,000
Zimbabwe	^e 710	800	310,000
Other countries ⁸	—	—	<u>3,300,000</u>
World total (rounded)	⁹ 107,000	⁹ 130,000	26,000,000

World Resources:⁶ Owing to continuing exploration, identified lithium resources have increased substantially worldwide and total about 98 million tons. Identified lithium resources in the United States—from continental brines, claystone, geothermal brines, hectorite, oilfield brines, and pegmatites—are 12 million tons. Identified lithium resources in other countries have been revised to 86 million tons. Identified lithium resources are distributed as follows: Bolivia, 21 million tons; Argentina, 20 million tons; Chile, 11 million tons; Australia, 7.9 million tons; China, 6.8 million tons; Germany, 3.2 million tons; Congo (Kinshasa), 3 million tons; Canada, 2.9 million tons; Mexico, 1.7 million tons; Czechia, 1.3 million tons; Serbia, 1.2 million tons; Russia, 1 million tons; Peru, 880,000 tons; Mali, 840,000 tons; Brazil, 730,000 tons; Zimbabwe, 690,000 tons; Spain, 320,000 tons; Portugal, 270,000 tons; Namibia, 230,000 tons; Ghana, 180,000 tons; Finland, 68,000 tons; Austria, 60,000 tons; and Kazakhstan, 50,000 tons.

Substitutes: Substitution for lithium compounds is possible in batteries, ceramics, greases, and manufactured glass. Examples are calcium, magnesium, mercury, and zinc as anode material in primary batteries; calcium and aluminum soaps as substitutes for stearates in greases; and sodic and potassic fluxes in ceramics and glass manufacture.

^eEstimated. W Withheld to avoid disclosing company proprietary data. — Zero.

¹Defined as production + imports – exports ± adjustments for Government and industry stock changes. Rounded to one significant digit to avoid disclosing company proprietary data.

²Lithium carbonate price assessments for spot and long-term contracts. Source: Benchmark Mineral Intelligence Ltd.

³Defined as imports – exports ± adjustments for Government and industry stock changes.

⁴See Appendix B for definitions.

⁵Units are kilograms, gross weight.

⁶See Appendix C for resource and reserve definitions and information concerning data sources.

⁷For Australia, Joint Ore Reserves Committee-compliant or equivalent reserves were 3.8 million tons.

⁸Other countries with reported reserves: Austria, Congo (Kinshasa), Czechia, Finland, Germany, Ghana, Mali, Mexico, Namibia, Serbia, and Spain.

⁹Excludes U.S. production.

MCS 2024

LITHIUM

U.S. Geological Survey, Mineral Commodity Summaries, January 2024

LITHIUM

(Data in metric tons, lithium content, unless otherwise specified)

Domestic Production and Use: Commercial-scale lithium production in the United States was from a continental brine operation in Nevada and from brine-sourced waste tailings of a Utah-based magnesium producer. Two companies produced a wide range of downstream lithium compounds in the United States from domestic or imported lithium carbonate, lithium chloride, and lithium hydroxide. Domestic production data were withheld to avoid disclosing company proprietary data.

Although lithium uses vary by location, global end uses were estimated as follows: batteries, 87%; ceramics and glass, 4%; lubricating greases, 2%; air treatment, 1%; continuous casting mold flux powders, 1%; medical, 1%; and other uses, 4%. Lithium consumption for batteries increased significantly in recent years because rechargeable lithium batteries have been used extensively in the growing market for electric vehicles, portable electronic devices, electric tools, and energy grid storage applications. Lithium minerals were used directly as mineral concentrates in ceramics and glass applications.

Salient Statistics—United States:

	<u>2019</u>	<u>2020</u>	<u>2021</u>	<u>2022</u>	<u>2023^e</u>
Production	W	W	W	W	W
Imports for consumption	2,620	2,460	2,640	3,270	3,400
Exports	1,660	1,200	1,870	2,440	2,300
Consumption, apparent ¹	W	W	W	W	W
Price, annual average-nominal, battery-grade lithium carbonate, dollars per metric ton ²	12,100	8,600	12,600	68,100	46,000
Employment, mine and mill, number	70	70	70	70	70
Net import reliance ³ as a percentage of apparent consumption	>25	>50	>25	>25	>25

Recycling: Construction of lithium battery recycling plants increased at a rapid pace. In 2023, about 40 companies in Canada and the United States and 50 companies in Europe recycled lithium batteries or planned to do so. Automobile companies and battery recyclers partnered to supply the automobile industry with a source of battery materials.

Import Sources (2019–22): Argentina, 51%; Chile, 43%; China, 3%; Russia, 2%; and other, 1%.

<u>Tariff:</u>	<u>Item</u>	<u>Number</u>	<u>Normal Trade Relations</u> <u>12–31–23</u>
	Lithium oxide and hydroxide	2825.20.0000	3.7% ad valorem.
	Lithium carbonate:		
	U.S. pharmaceutical grade	2836.91.0010	3.7% ad valorem.
	Other	2836.91.0050	3.7% ad valorem.

Depletion Allowance: 22% (domestic), 14% (foreign).

Government Stockpile: Not available.

Events, Trends, and Issues: Excluding U.S. production, worldwide lithium production in 2023 increased by 23% to approximately 180,000 tons from 146,000 tons in 2022 in response to strong demand from the lithium-ion battery market. Global consumption of lithium in 2023 was estimated to be 180,000 tons, a 27% increase from the revised consumption figure of 142,000 tons in 2022. However, concern of a short-term lithium oversupply, expiration of the Chinese Government's decade-long program of subsidies for electric vehicle (EV) purchases, and weaker-than-expected EV sales worldwide caused the price of lithium to decrease considerably in 2023.

Spot lithium carbonate prices in China [cost, insurance, and freight (c.i.f.)] decreased from approximately \$76,000 per ton in January to about \$23,000 per ton in November. For fixed contracts, the annual average U.S. lithium carbonate price was \$46,000 per ton in 2023, a decrease of 32% from that in 2022. Spot lithium hydroxide prices in China (c.i.f. China) decreased from approximately \$81,500 per ton in January to about \$22,500 per ton in November. Spodumene (6% lithium oxide) prices in Australia (free on board) decreased from approximately \$6,000 per ton in January to about \$2,500 per ton in November.

Seven mineral operations in Australia, one mineral tailings operation in Brazil, two brine operations each in Argentina and Chile, two mineral operations in Canada, five mineral and four brine operations in China, and one mineral operation in Zimbabwe accounted for the majority of world lithium production. Additionally, smaller operations in Argentina, Australia, Brazil, China, Portugal, the United States, and Zimbabwe also contributed to world lithium

LITHIUM

production. Owing to the rapid increase in demand of lithium in 2023, established lithium operations worldwide increased or were in the process of increasing production capacity.

In 2022, the U.S. Department of Energy selected 12 lithium-based projects funded with \$1.6 billion from the 2022 U.S. Bipartisan Infrastructure Law to support new commercial-scale domestic facilities to extract and process lithium, manufacture battery components, recycle batteries, and develop new technologies to increase U.S. lithium reserves. The 2022 U.S. Inflation Reduction Act added tax incentives to consolidate sourcing of battery materials and manufacturing of EVs to North American and U.S.-partner countries.

Lithium supply security has become a top priority for technology companies in Asia, Europe, and North America. Strategic alliances and joint ventures among technology companies and exploration companies continued to be established to ensure a reliable, diversified supply of lithium for battery suppliers and vehicle manufacturers. Brine-based lithium sources were in various stages of development or exploration in Argentina, Bolivia, Canada, Chile, China, and the United States; mineral-based lithium sources were in various stages of development or exploration in Australia, Austria, Brazil, Canada, China, Congo (Kinshasa), Czechia, Ethiopia, France, Finland, Germany, Ghana, India, Iran, Kazakhstan, Mali, Namibia, Nigeria, Peru, Portugal, Russia, Rwanda, Serbia, Spain, Thailand, Turkey, the United States, and Zimbabwe; lithium-clay sources were in various stages of development or exploration in Mexico and the United States.

World Mine Production and Reserves: Reserves for Argentina, Australia, Brazil, China, the United States, and “Other countries” were revised based on company and Government reports.

	Mine production		Reserves⁴
	<u>2022</u>	<u>2023^e</u>	
United States	W	W	1,100,000
Argentina	6,590	9,600	3,600,000
Australia	74,700	86,000	⁵ 6,200,000
Brazil	^e 2,630	4,900	390,000
Canada	^e 520	3,400	930,000
Chile	38,000	44,000	9,300,000
China	^e 22,600	33,000	3,000,000
Portugal	^e 380	380	60,000
Zimbabwe	^e 1,030	3,400	310,000
Other countries ⁶	—	—	<u>2,800,000</u>
World total (rounded)	<u>7146,000</u>	<u>7180,000</u>	<u>28,000,000</u>

World Resources:⁴ Owing to continuing exploration, measured and indicated lithium resources have increased substantially worldwide and total about 105 million tons. Measured and indicated lithium resources in the United States—from continental brines, claystone, geothermal brines, hectorite, oilfield brines, and pegmatites—are 14 million tons. Measured and indicated lithium resources in other countries have been revised to 91 million tons. Resources are distributed as follows: Bolivia, 23 million tons; Argentina, 22 million tons; Chile, 11 million tons; Australia, 8.7 million tons; China, 6.8 million tons; Germany, 3.8 million tons; Canada, 3 million tons; Congo (Kinshasa), 3 million tons; Mexico, 1.7 million tons; Czechia, 1.3 million tons; Serbia, 1.2 million tons; Peru, 1 million tons; Russia, 1 million tons; Mali, 890,000 tons; Brazil, 800,000 tons; Zimbabwe, 690,000 tons; Spain, 320,000 tons; Portugal, 270,000 tons; Namibia, 230,000 tons; Ghana, 200,000 tons; Finland, 68,000 tons; Austria, 60,000 tons; and Kazakhstan, 50,000 tons.

Substitutes: Substitution for lithium compounds is possible in batteries, ceramics, greases, and manufactured glass. Examples are calcium, magnesium, mercury, and zinc as anode material in primary batteries; calcium and aluminum soaps as substitutes for stearates in greases; and sodic and potassic fluxes in ceramics and glass manufacture.

^eEstimated. W Withheld to avoid disclosing company proprietary data. — Zero.

¹Defined as production + imports – exports ± adjustments for industry stock changes.

²Lithium carbonate price assessments for spot and long-term contracts. Source: Benchmark Mineral Intelligence Ltd.

³Defined as imports – exports ± adjustments for industry stock changes.

⁴See Appendix C for resource and reserve definitions and information concerning data sources.

⁵For Australia, Joint Ore Reserves Committee-compliant or equivalent reserves were 4.8 million tons.

⁶Other countries with reported reserves: Austria, Congo (Kinshasa), Czechia, Finland, Germany, Ghana, Mali, Mexico, Namibia, Serbia, and Spain.

⁷Excludes U.S. production.

MCS 2025

LITHIUM

U.S. Geological Survey, Mineral Commodity Summaries, January 2025

LITHIUM

(Data in metric tons, lithium content, unless otherwise specified)

Domestic Production and Use: Commercial-scale lithium production in the United States was from a continental brine operation in Nevada. Owing to lower lithium prices in 2024, commercial production from the brine-sourced waste tailings of a Utah-based magnesium producer was idled. Two companies produced a wide range of downstream lithium compounds in the United States from domestic or imported lithium carbonate, lithium chloride, and lithium hydroxide. Domestic production data were withheld to avoid disclosing company proprietary data.

Although lithium uses vary by location, global end uses were estimated as follows: batteries, 87%; ceramics and glass, 5%; lubricating greases, 2%; air treatment, 1%; continuous casting mold flux powders, 1%; medical, 1%; and other uses, 3%. Lithium consumption for batteries increased significantly owing to the use of rechargeable lithium batteries in the growing market for electric vehicles (EVs), portable electronic devices, electric tools, and energy grid storage applications. Lithium minerals were used directly as mineral concentrates in ceramics and glass applications.

Salient Statistics—United States:

	2020	2021	2022	2023	2024^e
Production	W	W	W	W	W
Imports for consumption	2,460	2,640	3,260	3,390	3,300
Exports	1,200	1,870	2,440	1,960	1,700
Consumption, apparent ¹	W	W	W	W	W
Price, annual average-real, battery-grade lithium carbonate, dollars per metric ton ²	10,100	14,200	71,100	41,300	14,000
Employment, mine and mill, number	70	70	70	70	70
Net import reliance ³ as a percentage of apparent consumption	>50	>25	>25	>50	>50

Recycling: Construction of lithium battery recycling plants continued throughout 2024. Automobile companies and battery recyclers partnered to supply the automobile industry with a source of battery materials. In October, the U.S. Department of Energy announced \$44.8 million in funding from the U.S. Bipartisan Infrastructure Law for eight projects that will help lower EV battery recycling costs, with the long-term goal of lowering vehicle costs.

Import Sources (2020–23): Chile, 50%; Argentina, 47%; and other, 3%.

Tariff:	Item	Number	Normal Trade Relations 12–31–24
	Lithium oxide and hydroxide	2825.20.0000	3.7% ad valorem.
	Lithium carbonate:		
	U.S. pharmaceutical grade	2836.91.0010	3.7% ad valorem.
	Other	2836.91.0050	3.7% ad valorem.

Depletion Allowance: 22% (domestic), 14% (foreign).

Government Stockpile: Not available.

Events, Trends, and Issues: Excluding U.S. production, worldwide lithium production in 2024 increased by 18% to approximately 240,000 tons from 204,000 tons in 2023 in response to strong demand from the lithium-ion battery market, high lithium prices from 2021 to early 2023, and an increase in global lithium production capacity. Global consumption of lithium in 2024 was estimated to be 220,000 tons, a 29% increase from revised consumption of 170,000 tons in 2023. Concern about a short-term lithium oversupply and weaker-than-expected EV sales worldwide during the first half of 2024 caused the price for lithium to decrease considerably throughout the year. Owing in part to incentives and discounts, EV sales in the third quarter of 2024 saw considerable growth in Canada, China, and the United States.

Spot lithium carbonate prices in China [cost, insurance, and freight (c.i.f.)] decreased from approximately \$14,500 per ton in January to approximately \$9,400 per ton in November. For fixed contracts, the annual average U.S. lithium carbonate price was \$14,000 per ton in 2024, a decrease of 66% from that in 2023. Spot lithium hydroxide prices in China [free on board (f.o.b.)] decreased from approximately \$17,000 per ton in January to approximately \$9,900 per ton in November. Spodumene (6% lithium oxide) prices in Australia (f.o.b.) decreased from approximately \$1,250 per ton in January to approximately \$730 per ton in November.

Four brine operations in Argentina, nine mineral operations in Australia, one mineral tailings operation in Brazil, two mineral operations in Canada, two brine operations Chile, seven mineral and five brine operations in China, and four mineral operations in Zimbabwe accounted for the majority of world lithium production. Additionally, smaller

Prepared by **Brian W. Jaskula [(703) 648–4908, bjaskula@usgs.gov]**

LITHIUM

operations in Australia, Brazil, China, Namibia, Portugal, and the United States also contributed to world lithium production. Despite many lithium projects being postponed or cancelled in 2024 owing to low prices, significant production capacity expansions occurred in Argentina, Chile, China, and Zimbabwe.

In 2024, the U.S. Department of Energy announced \$3 billion in funding across 25 projects through the U.S. Bipartisan Infrastructure Law to support new commercial-scale domestic facilities to extract and process lithium and other critical minerals, manufacture key battery components, recycle batteries, support next-generation battery manufacturing, and develop new technologies to increase U.S. lithium reserves.

Lithium supply security has become a priority for technology companies in Asia, Europe, and North America. Strategic alliances and joint ventures among technology companies and exploration companies continued to be established to ensure a reliable, diversified supply of lithium for battery suppliers and vehicle manufacturers. Brine-based lithium sources were in various stages of development or exploration in Argentina, Bolivia, Canada, Chile, China, and the United States; mineral-based lithium sources were in various stages of development or exploration in Australia, Austria, Brazil, Canada, China, Congo (Kinshasa), Czechia, Ethiopia, Finland, France, Germany, Ghana, India, Iran, Kazakhstan, Mali, Namibia, Nigeria, Peru, Portugal, Russia, Rwanda, Serbia, Spain, Thailand, Turkey, the United States, and Zimbabwe; lithium-clay sources were in various stages of development or exploration in Mexico and the United States.

World Mine Production and Reserves: Reserves for Argentina, Australia, Canada, the United States, and Zimbabwe were revised based on company and Government reports.

	Mine production		Reserves ⁴
	2023	2024 ^e	
United States	W	W	1,800,000
Argentina	8,630	18,000	4,000,000
Australia	91,700	88,000	⁵ 7,000,000
Brazil	^e 5,260	10,000	390,000
Canada	^e 3,240	4,300	1,200,000
Chile	41,400	49,000	9,300,000
China	^e 35,700	41,000	3,000,000
Namibia	^e 2,700	2,700	14,000
Portugal	^e 380	380	60,000
Zimbabwe	^e 14,900	22,000	480,000
Other countries ⁶	—	—	2,800,000
World total (rounded)	⁷ 204,000	⁷ 240,000	30,000,000

World Resources:⁴ Owing to continuing exploration, measured and indicated lithium resources have increased substantially worldwide and total about 115 million tons. Measured and indicated lithium resources in the United States—from continental brines, claystone, geothermal brines, hectorite, oilfield brines, and pegmatites—are 19 million tons. Measured and indicated lithium resources in other countries have been revised to 96 million tons. Resources are distributed as follows: Argentina, 23 million tons; Bolivia, 23 million tons; Chile, 11 million tons; Australia, 8.9 million tons; China, 6.8 million tons; Canada, 5.7 million tons; Germany, 4 million tons; Congo (Kinshasa), 3 million tons; Mexico, 1.7 million tons; Brazil, 1.3 million tons; Czechia, 1.3 million tons; Mali, 1.2 million tons; Serbia, 1.2 million tons; Peru, 1 million tons; Russia, 1 million tons; Zimbabwe, 860,000 tons; Spain, 320,000 tons; Portugal, 270,000 tons; Namibia, 230,000 tons; Ghana, 200,000 tons; Austria, 60,000 tons; Finland, 55,000 tons; and Kazakhstan, 45,000 tons.

Substitutes: Substitution for lithium compounds is possible in batteries, ceramics, greases, and manufactured glass. Examples are calcium, magnesium, mercury, and zinc as anode material in primary batteries; calcium and aluminum soaps as substitutes for stearates in greases; and sodic and potassic fluxes in ceramics and glass manufacture.

^eEstimated. W Withheld to avoid disclosing company proprietary data. — Zero.

¹Defined as production + imports – exports ± adjustments for industry stock changes.

²Lithium carbonate price assessments for spot and long-term contracts. Source: Benchmark Mineral Intelligence Ltd.

³Defined as imports – exports ± adjustments for industry stock changes.

⁴See Appendix C for resource and reserve definitions and information concerning data sources.

⁵For Australia, Joint Ore Reserves Committee-compliant or equivalent reserves were 4.8 million tons.

⁶Other countries with reported reserves include Austria, Congo (Kinshasa), Czechia, Finland, Germany, Ghana, Mali, Mexico, Serbia, and Spain.

⁷Excludes U.S. production.